Ship Hull Biofouling Detection using YOLOv8

Project Title

Ship Hull Biofouling Detection using YOLOv8 and Streamlit

Project Objective

To automatically detect marine growth and biofouling on ship hull surfaces using a custom YOLOv8 object detection model and provide image and video-based inspection through a Streamlit web application.

Dataset Preparation

Collected ship hull inspection videos, extracted frames, annotated images, and organized the dataset into YOLO format with train, validation, and test sets.

Workflow

1. Data Collection
2. Frame Extraction
3. Data Annotation
4. YOLO Dataset Preparation
5. YOLOv8 Training
6. Model Evaluation
7. Image Detection
8. Video Detection
9. Streamlit Deployment

Technologies Used

Python, YOLOv8, OpenCV, Streamlit, FFmpeg, PIL, Ultralytics

Classes Used

clean_hull, marine_growth, heavy_marine_growth

Model Training

Trained a custom YOLOv8 model and generated best.pt weights for inference.

Image Detection Module

Users can upload ship hull images. The model predicts marine growth, draws bounding boxes, and displays confidence scores.

Video Detection Module

Users can upload MP4/AVI/MOV videos. YOLO processes all frames and generates an annotated output video.

Challenges Faced

Handling large videos, Streamlit video playback issues, FFmpeg integration, output video conversion, and model inference optimization.

Solutions Implemented

Used FFmpeg to convert AVI outputs to MP4 for browser compatibility and integrated YOLO inference directly into the Streamlit interface.

Project Outcome

Successfully developed an end-to-end AI-powered marine biofouling detection system capable of analyzing both images and videos.

Future Enhancements

Cloud deployment, detection analytics dashboard, downloadable reports, detection statistics, and real-time inspection support.